



COMBIJET JE60-500

HIGH PRESSURE
WATER JET CLEANER
USER & PARTS MANUAL
V4.8



Thank you for choosing our water blaster. For your safety and correct operation, please carefully read the information and instructions in this manual before use.

Combijet has no responsibility or liability regarding misuse of the product with third party products.

Combijet warranty's is limited to repair or replacement of faulty items at the time of manufacture.

Model: JE60-500**Technical details:**

Working Pressure: 500 bar / 7250 PSI

Flow-rate (@50Hz): 16 lpm / 4.2 GPM

Flow-rate (@60Hz): 19 lpm / 5.0 GPM

Electrical Specifications:

Motor: 20 HP / 15 kW

Voltage: 380-440V

Amperage: 27 A

Dimensions LxWxH: 1030 x 630 x 920 mm

Weight: 265 kg



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HIGH PRESSURE JET-CLEANER JE60-500

USER AND PARTS MANUAL

V4.8

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WARNING & SAFETY PRECAUTIONS

WARNINGS

- 1  Make sure to **ground** the Unit before you start any further procedure.
- 2  **WARNING:** This appliance is for outdoor use only.
- 3  **WARNING:** Always disconnect the electricity and water supplies on completion of every job.
- 4  **WARNING:** Do not use the machine if a supply cord or important parts of the machine are damaged, e.g. safety devices, high pressure hoses, trigger gun.
- 5  **WARNING:** This appliance has been designed for use with the detergent provided or the detergent specified by the manufacturer, such as a neutral shampoo based on biodegradable anionic surface-active detergents. The use of other detergents or chemicals may put the appliance's safety at risk.
- 6  **WARNING:**
 - a) Do not use the machine within range of persons unless they wear protective clothing.
 - b) Do not allow other people or animals within a range of 5m when operating the machine.
 - c) Always wear suitable protective clothing in order to protect yourself against ricocheting parts.
 - d) Do not touch the plug and the appliance with wet hands or when barefoot.
 - e) Wear safety goggles and non-slip rubber footwear.
- 7  **WARNING:** Do not direct the nozzle toward mechanical parts containing lubricant grease, as the grease will dissolve and spread over the surrounding ground.
- 8 Vehicle tyres/tyre valves may only be cleaned from a minimum distance of 30 cm, otherwise the vehicle tyre/tyre valve could be damaged by the high-pressure jet. The first indication of this is a discoloration of the tyre. Damaged vehicle tyres/tyre valves can be deadly dangerous.
- 9  **WARNING:** High pressure jets can be dangerous if subject to misuse. The jet must not be directed at persons, animals, live electrical equipment or the machine itself.
- 10  **WARNING:** High pressure hoses, fittings and couplings are important for the safety of the machine. Use only hoses, fittings and couplings recommended by the manufacturer. (It is extremely important to protect these components against damage by avoiding their improper use and protecting them against bending, knocks and scratches).
- 11  **WARNING:** Switch the appliance off completely (master switch ON/OFF) whenever it is left unattended.
- 12  **WARNING:** Every machine is tested in its operating conditions, so it is normal for a few drops of water to be left inside it.
- 13  **WARNING:** Take care not damage the electric cable. If the supply cord is damaged, it must be replaced by the manufacturer, its service agent or similarly qualified persons in order to avoid a hazard
- 14  **WARNING:** Appliance with fluid under pressure. Keep a firm hold on the gun and be prepared to take the kickback pressure. Only use the high pressure nozzle provided with the appliance.
- 15  **WARNING:** This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.

- 16  **WARNING:** Children should be supervised to ensure that they do not play with the appliance.
- 17  **WARNING:** Never start the high-pressure cleaner without first completely unwinding the high-pressure hose.
- 18  **WARNING:** When winding and unwinding the pipe, take care not to cause the high pressure cleaner to overturn.
- 19  **WARNING:** Before unwinding or winding the hose, switch the machine off and release the pressure in the hose itself (switching off).
- 20  **WARNING:** Risk of explosion — Do not spray flammable liquids.
- 21  **WARNING:** To ensure machine safety, use only original spare parts from the manufacturer or approved by the manufacturer.
- 22  **WARNING:** Do not direct the jet against yourself or others in order to clean clothes or footwear.
- 23  **WARNING:** High pressure cleaners shall not be used by children or untrained personnel.
- 24  **WARNING:** Water that has flown through backflow preventers is considered to be non-potable.
- 25  **WARNING:** the machine shall be disconnected from its power source, by removing the plug from the socket-outlet, during cleaning or maintenance and when replacing spare parts
- 26  **WARNING:** If an extension cord is used, the plug and socket must be of watertight construction.
- 27  **WARNING:** Inadequate extension cords can be dangerous.
- 28  **WARNING:** It is absolutely forbidden to use the machine in potentially explosive environments or areas.
- 29  **WARNING:**
- 29.1 NEVER EVER TOUCH THE MACHINE WITH WET HANDS OR BARE FEET.
- 29.2 NEVER PULL THE POWER SUPPLY CABLE OR MACHINE IN ORDER TO DISCONNECT THE PLUG FROM THE SOCKET.
- 29.3 IF DURING FUNCTIONING THE POWER SUPPLY RUNS OUT, FOR SAFETY REASONS, TURN THE MACHINE OFF.

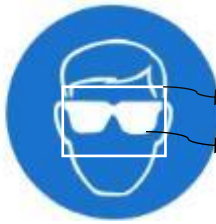
SAFETY DEVICES

-  **WARNING:** the gun is fitted with a safety catch. Whenever use of the machine is interrupted it is important to operate the safety catch to prevent accidental activation of the jet.
 - Safety features: gun equipped with safety locking device, appliance equipped with (Class I) overload cut-out, pump equipped with by-pass valve or shutdown device.
 - The safety button on the gun is not there to lock the lever during operation, but to prevent its accidental operation.
-  **WARNING:** The appliance is equipped with a motor protection device: in case of device intervention, wait some minutes or, in alternative, disconnect and re-connect the product to the electric system. In case this problem repeats again or if the product does not start again, take the product to the nearest After Sales Service Point

STABILITY

-  **WARNING:** The machine must be always kept on horizontal bases, in a safe and stable way.

PROTECTION GEAR



Visor

Goggles

EYE PROTECTION

Operators must wear visors or goggles to guard against spray and flying debris. A combination of both goggles and visor is recommended. It will protect the eyes and the face during both water-blasting as well as abrasive water-blasting.



HEAD PROTECTION

Safety helmets must be worn at all time by personnel within the work area.

Helmet material must be able to withstand mechanical shock exceeding 10G in 8ms without fracturing.



HAND PROTECTION

Shear proof gloves must be worn by the operator at all times.

A glove combination of a cloth inner lining and a water tight outer layer is preferable.



FOOT PROTECTION

Safety boots with steel toecaps minimum 0.5mm (0.02") thick must be worn. The toecap must cover at least 30% of the footwear length. Basic safety footwear must also be equipped with metatarsal guards to protect the instep. Safety boots are available in numerous designs.



HEARING PROTECTION

Operators and other personnel exposed to noise levels of more than 90-dBa for more than 1 hour must wear suitable ear protection. Ear plugs or ear muffs are usually sufficient.



PROTECTIVE CLOTHING

Waterproof clothes protect the operator only from spray and flying debris. They do NOT deflect direct jet impact.

Therefore, an operator must take care never to point a water jet either at themselves or other personnel.

MEDICAL EMERGENCY INFORMATION

Immediate medical attention should be given to personnel who sustain equipment related injuries while operating the system. In such cases, it is vital that medical personnel be apprised of all facts relevant to such injuries. Therefore, all operating personnel should be provided with waterproof emergency medical alert tags or cards, describing the nature of their work and the possibility of injury inherent in the use of water jetting equipment. The below example of a standard card can be printed out or photo copied, bend, laminated and used as medical alert tag.

MEDICAL ALERT

This card is to be carried by personnel working with high-pressure water jetting equipment. Obtain medical treatment immediately for ANY high-pressure water-jet injuries.



This person has been working with water jetting at pressures up to 55,000 psi (374Mpa, 3,740bar, 3,867 kg/cm²) with a jet velocity of up to 3,000 fps (914 mps). This should be taken into account during diagnosis. Unusual infections with microaerophilic organisms occurring at lower temperatures have been reported. These may be gram-negative pathogens such as found in sewage. Bacterial swabs and blood cultures may therefore be helpful. A local poison control centre should be contacted for additional treatment information and advise.



member of:  **Demetriades group**

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MACHINE NAMEPLATE

		Combijet Demetriades Group P.O.Box 59617, 4011 Limassol, Cyprus Tel.: +357 25755866 - Fax: +357 25725940 info@combijet.com		
MODEL / TYPE		JE60-500		
SERIAL No _____				
WORKING PRESSURE		500	Bar	7100 Psi
FLOWRATE		16/18	L/Min	4/4.5 Gpm
POWER RATING		20	Hp	15 Kw
380/440 Volt		23	Amp Power	1450/1750 Rpm
www.combijet.com				

SET-UP & USE

A. Place the machine in a safe location and secure the unit with leashing belts.

B. Freshwater Connection

Prepare the water supply for the blaster, by attaching a ¾" hose to the supplied low-pressure inlet coupling on the side of the pump. Always use the inlet connection on the pump for supplying water.

Water supply collection

Water supply from the water main

- Connect a supply hose (not included) to the water inlet of the unit and to the water supply.
- Open the water supply.



WARNING: machine not suitable for connection to the potable water mains.

The cleaner can only be connected to the water mains if the water main is separated by a backflow preventer. Make sure that the hose is at least Ø19mm-3/4" inch and that it is reinforced.



WARNING: Water that has flown through backflow preventers is considered to be non-potable.

IMPORTANT: Only clean or filtered water should be used for intake. The delivery of the water intake tap should be equal to the double of the maximum pump range.

- Minimum delivery rate: 30 l/min.
- Maximum intake water temperature: 40°C
- Inlet water pressure: 1-1.5Mpa

Place the cleaner as close to the water supply system as possible.

Procedure for water supply from an open container

- 1) Unscrew the high-pressure line at the high pressure outlet of the unit.
- 2) Switch on the unit and let it run until water free of bubbles emerges at the high-pressure outlet.
- 3) Switch off the unit and screw on the high pressure hose again.

If the machine will be inactive for a certain period then perform the procedure again as air maybe trapped in the system.

Failure to comply with the above conditions causes serious mechanical damage to the pump and the loss of warranty cover.

C. High Pressure Hose Connection

Mount the ½" high pressure hose to the blaster by attaching the swivel of the hose to the outlet fitting of the pump head. Please use 2 x 27mm spanners / wrenches to tighten the connection. Additional high-pressure hoses supplied with the machine should be mounted now by using ½" nipples male – male to connect the hoses together. Please use 2 x 27mm spanners / wrenches to tighten the connection

D. Trigger Gun Connection

Mount the trigger gun with lance to the high-pressure hose. Please use 2 x 27mm spanners / wrenches to tighten the connection.

Mount the correct nozzle at the end of the lance.

E. Electrical Power Connection

Connect the power cable CEE plug to an earthed 3 phase 32 Amp power socket.

- The machine is very important to be earthed after the installation.
- The appliance's electrical connection must comply with the IEC 60364-1 standard.

IMPORTANT: Before connecting the appliance, make sure that the nameplate data are the same as those of the electrical mains and

- Never use the appliance in ambient temperatures below 0° C if it is equipped with a PVC (H VV-F) cable.



WARNING: Inadequate extension cords can be dangerous.



WARNING: If an extension cord is used, the plug and socket must be of watertight construction. Information for supply may also be obtained from the plug manufacturer.

Cable Rated Voltage: 380-440V

Standard Cable Size: 4G x 6mm²

For extension of power cord please follow the following guidelines:

Cable Length (m)	Cable Size
0-20	4G x 6mm ²
20-50	4G x 10mm ²
50-100	4G x 16mm ²

Abbreviations:

- a. Number of wires: Four (4) – 3xL + 1xE
 L: stands for Live Wires
 E: stands for Earth Wire
- b. Cross Sectional Area of the wire.

F. Machine is now ready for Operation

Operation of the machine:

- ✓ Ensure that the water supply is open
- ✓ Turn the main power switch to ON. Press the green START button – machine starts. Allow the machine few seconds for the automatic star – delta switch to turn over.
- ✓ Get a firm grip with both hands on the trigger gun handle and lance handle. Press the trigger and start blasting. Always be aware of other crew members in your working area – **Safety is first priority.**

NOTE: Do NOT let the machine run idle more than 10 minutes, as it may cause damage to the pump. (Water heats up).

Shut-Down of the Machine:

- ✓ Press the red STOP button – machine stops.
- ✓ Squeeze the trigger gun to release the pressure inside the high-pressure hose.
- ✓ Turn off the water supply.

○ On-board vessels

Position the machine behind the super structure or on the poop deck, securely fastened. Engage the built-in brake when running the machine.

▪ 1ST Start-up

When starting up for the first time, or after a long period of use, connect the intake line only for a few minutes so that any dirt is vented from the delivery outlet.

DAILY INSPECTION

CHECK LIST

- Check the strainer in the water-tank.
- Clean if necessary.
- Check the oil level on the crankcase of the pump.
- Fill up if necessary.
- Check for air-bubbles in the suction hose between tank and pump.
- If there are bubbles then tighten the hose-clips.
- Check high pressure connections for leakage.
- NEVER tighten connections under pressure.
- Check high pressure hoses for wear and cracks.
- Replace if necessary.
- Check the power cord for wear and cracks.
- Replace if necessary.
- Check the air intake grill of the electric motor.
- Clean for dust if necessary.

NOTICES:

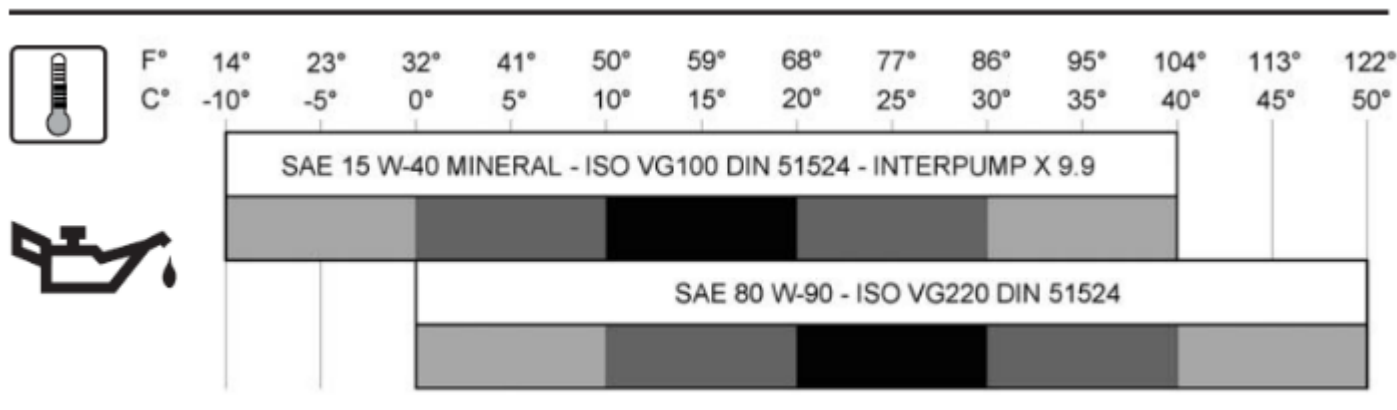
- To prevent damage to hoses and electrical cord, avoid letting the high-pressure hoses rub against sharp edges for example manholes or getting squeezed in door openings etc. Replace if necessary.
- Inlet water temperature must be below 60°C (140°F)

MAINTENANCE

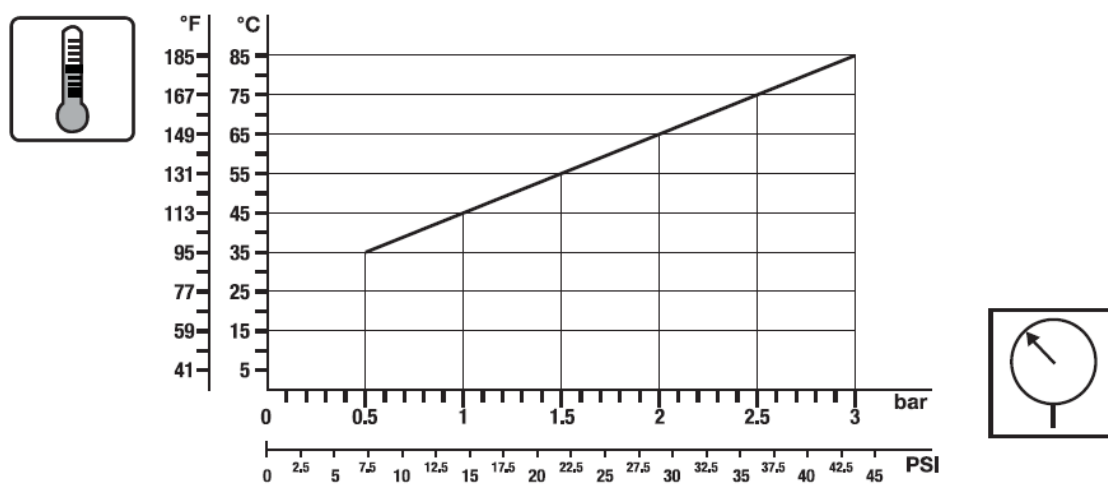
CHECKING THE PUMP OIL LEVEL

Check the oil level periodically. The oil must be changed for the first time after 50 working hours, with subsequent changes every 500 hours.

Recommended oils: SAE 20/30, 15W-40, 20W-40



MINIMUM FEED PRESSURE



CHECKING THE WATER INTAKE FILTER

Inspect and clean the water intake filter periodically. This is important to prevent damage to the operation of the pump.

CHECKING THE SPRAY NOZZLE

Replace the nozzle when the pressure drops below the normal values.

DESCALING

The descaling procedure must be carried out periodically with specific products. The interval depends on the hardness of the water. Pour one liter of product for every 15 liters of water into a drum containing at least 30 liters of water. Disconnect the gun from the high-pressure line and place its free end in the drum, in order to form a closed circuit, and draw the solution into the machine for at least 10 minutes. The delivery end of the line should discharge into a canvas or mesh bag so that the scale removed is not returned to the circuit. Then restore the normal connection and rinse with plenty of cold water. This operation should be carried out by our after-sales service, since the descaling product may cause wear of components. Comply with the regulations in force when disposing of the descaling agent.

CLEANING

A part of keeping the machine and its accessories in good working condition is keeping it clean. But NEVER clean the machine itself with high-pressure. Do not use solvents.

STORAGE

Inactivity for long periods

If the machine is to be inactive for long periods, keep the unit frost free, and flush the pump thoroughly with non-alcoholic antifreeze fluid.

Filling the pump with an anti-corrosion emulsion or anti-freeze solution by using an external diaphragm pump

- a) Close the filter draining, if open.
- b) Be sure that the connecting pipe is clean, spread with grease and connect it to the high pressure outlet port.
- c) Fit a suction hose to the membrane pump. Open the pump suction connection and fit hose between it and the membrane pump.
- d) Fill the container with the solution / emulsion.
- e) Put the free extremities of the suction pipe and the high pressure outlet pipe inside the container
- f) Start up the diaphragm pump
- g) Pump the emulsion until it comes out of the high pressure outlet pipe
- h) Continue pumping for at least another minute; if needed, the emulsion can be re-enforced, by adding for example Shell Donax
- i) Stop the pump, remove the pipe from the suction connection and close it with a plug
- j) Remove the pipe from the high pressure outlet port. Clean, grease and plug both connections and the pipes.

Pipes

- a) Before greasing and protecting the pipes according the previously described procedure, dry the connections using compressed air
- b) Cover with polyethylene
- c) Do not wrap them too tightly; be sure there is no folding

Starting after long period of inactivity

If the machine is started up for the first time after a long period of time, before starting operation check the oil level and the valves, and follow the start-up procedure described

PRECAUTIONS AGAINST FREEZING

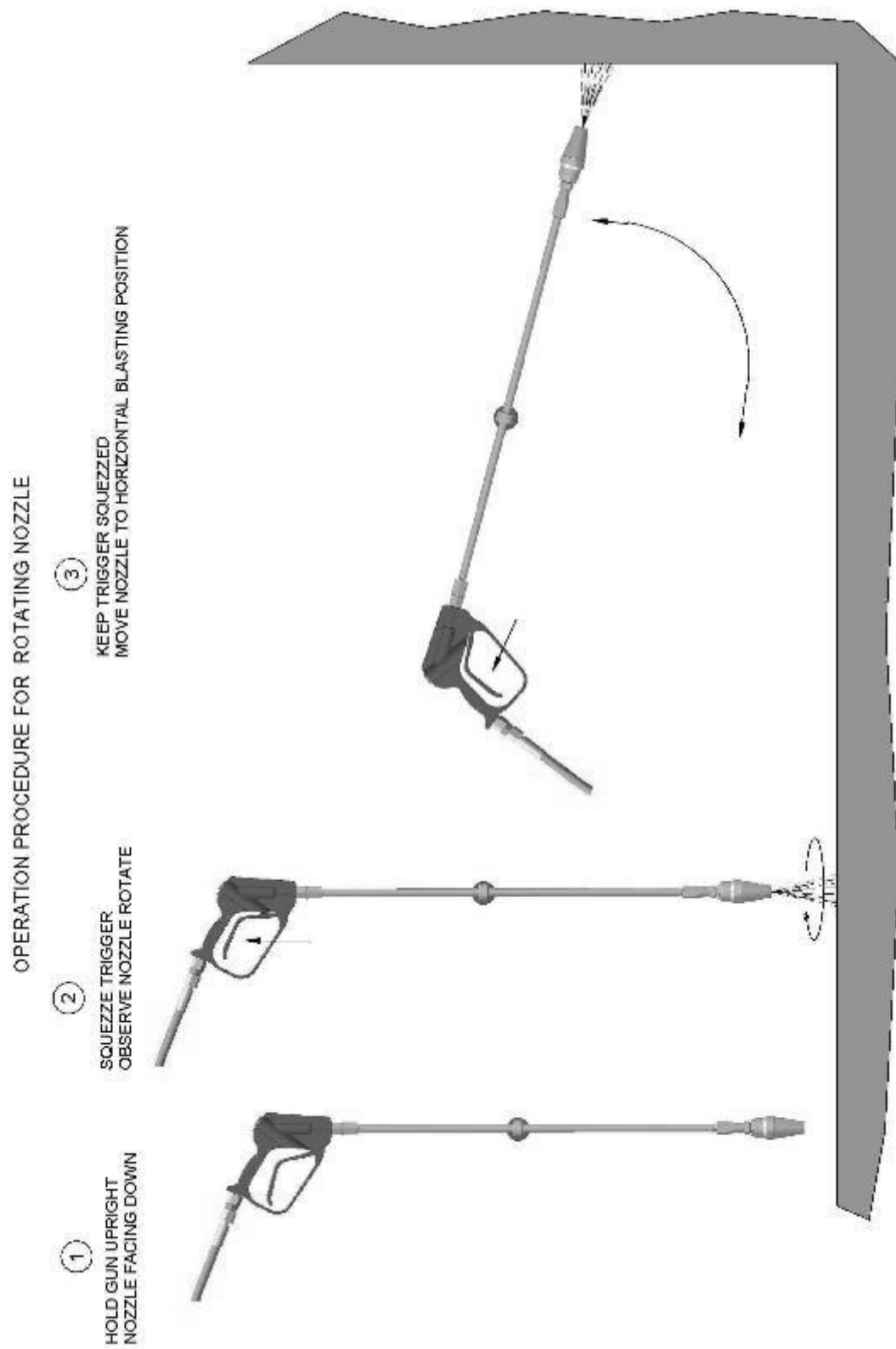


In areas and periods of the year where there is risk of freezing, follow the instructions indicated above in STORAGE section.

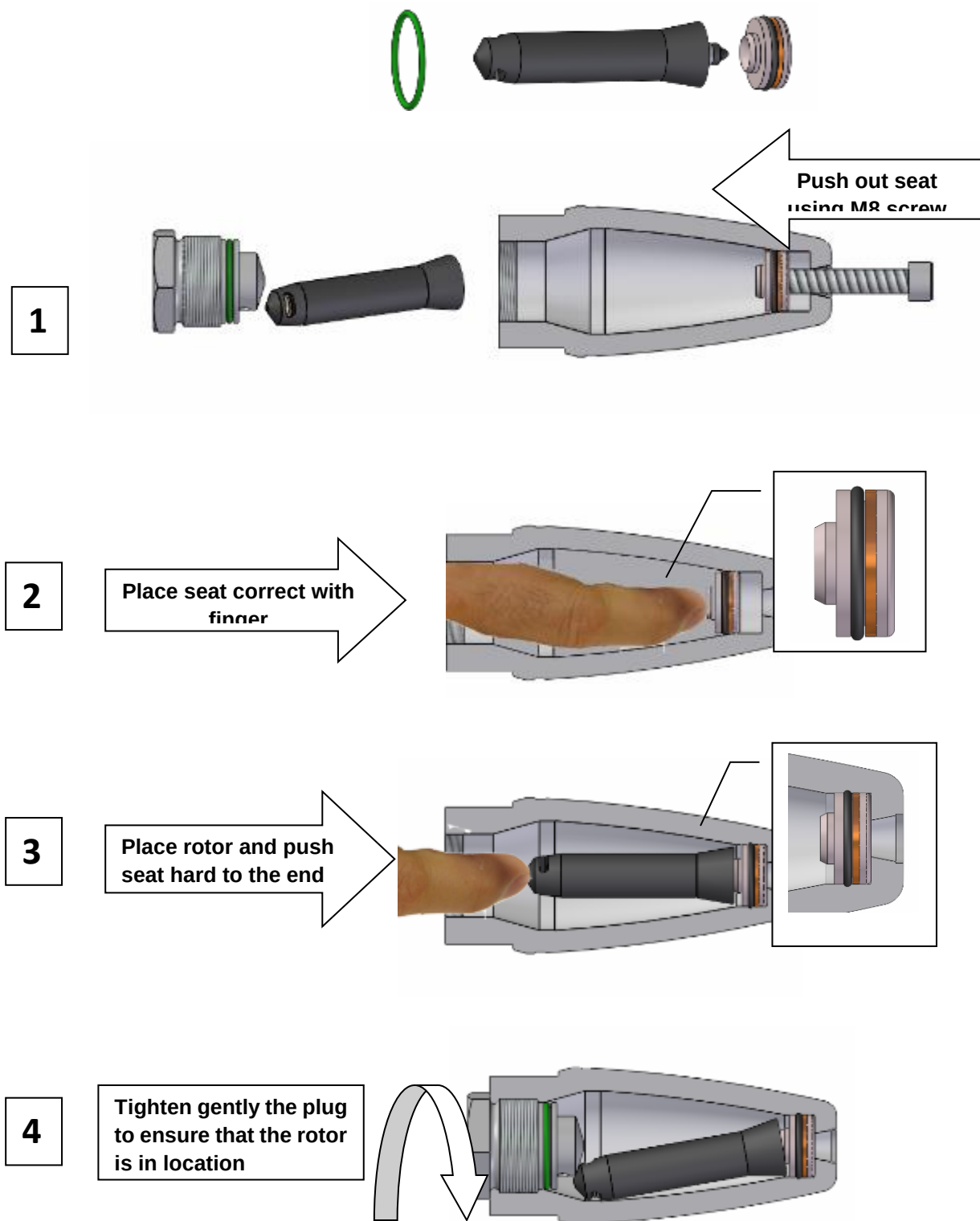


In the presence of ice, in no case must the pump be started until the entire circuit has been perfectly thawed out; not complying with this indication may cause serious damage to the pump.

OPERATION PROCEDURE FOR ROTATING NOZZLE



ROTATING NOZZLE REPAIR PROCEDURE



WET SAND BLASTING ACCESSORY USE & MAINTENANCE

BEFORE YOU BEGIN

WARNING

To reduce the risk of injury, always protect eyes and face with goggles and mask, and hands and arms with heavy work gloves when spraying abrasive materials.

- Always point the sand nozzle downward when not spraying. This prevents water from entering the sand supply.
- Always be sure the sand hose is dry before using.
- Keep the sand covered to prevent the overspray from wetting the sand.
- Do not allow small pieces of the sand bag to fall into the sand supply. A small piece of paper could prevent the flow of sand.

OPERATION

- a) Place the sand probe into the sand supply. Locate the sand in a dry location and protect the sand from water contact.
- b) Wet sand will prevent sand suction.
- c) Quick connect the sandblaster assembly to the end of the lance where quick connect nozzles normally attach.
- d) Turn off your water supply and pull the trigger to purge the system of air.
- e) Start pressure washer following the instruction manual for your washer.
- f) Become comfortable with the sandblaster, and test its operation on a scrap piece of media prior to use. Observe the impact of the sand at different angles and distances from the surfaces.

USE & MAINTENANCE

With the aim of guarantee a correct function and to avoid damages to people or objects, we recommend to strictly follow the instructions reported on this manual.

SET UP AND CONNECTION



WET SANDBLASTING NOZZLE ACCESSORY



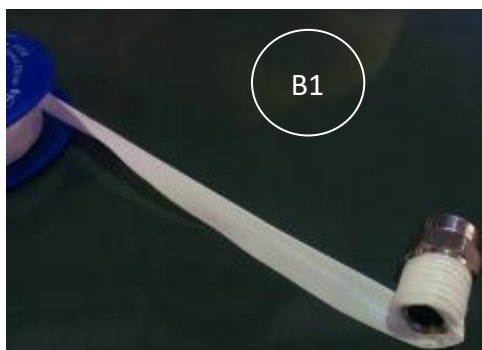
NOZZLE



ADAPTOR



- A. Remove the inlet fitting (4) from the main body (2), leaving the nut (3) on it.



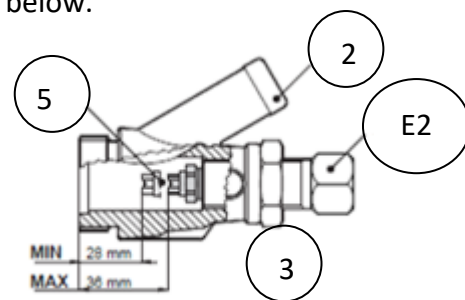
- B. Use TPFE tape on the nozzle (5), screw it on the inlet fitting (4) shown above (B3).



- C. Use TPE tape on the adaptor (6) and then screw it on the inlet fitting (4)



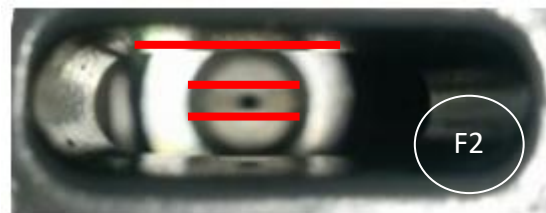
- D. Screw the inlet fitting (4) on the main body (3) as shown below.



- E. Unscrew the main body cap (1) as shown above and adjust the nozzle height as shown in figure E2 and lock the position by tightening the nut on the main body (2).



Note: Now the nozzle (3) outlet in the main body (2) is visible and also the main body cap (1) front outlet slot.



- F. Screw the main body cap (1) on the main body (2) and adjust the slot to be parallel with the nozzle slot by adjusting the main body cap.

IMPORTANT NOTES:

- Slots must be parallel otherwise the nozzle will not work properly
- Screwdriver use for nozzle adjustment could harm the nozzle outlet and that will lead to its defectiveness.

For maintenance purposes

- Unscrew the inlet fitting (4) from the main body (2)
- Unscrew nozzle (5) from the inlet fitting (4) and clean it if it's blocked.
- Then follow steps D, E & F to assemble the wet sand blasting accessory.

MACHINE TROUBLESHOOTING & REPAIR

FAULT	CAUSE	REPAIR
When switch “ON” the machine will not start working	• Plug not properly inserted in socket.	• Fit plug in power socket correctly.
	• Electric socket is faulty.	• Have electrical socket checked.
	• The main voltage is insufficient.	• Have a check made on the suitability of the mains system.
	• The pump is stuck. • Gauge of extension lead not sufficient.	• Turn switch to ON while keeping nozzle trigger pressed; if problem persists, contact an authorised service center.
The motor stops suddenly	• Thermal safety switch has tripped	• Leave motor to cool for a few minutes. If the problem persists, contact an authorised service center.
	• Cable extension seems to be inadequate.	
The pump doesn’t reach the necessary pressure	• Insufficient flow.	• Check that the delivery rate is 30% higher than the value stated on the appliance’s technical data plate.
	• Water inlet filter is clogged.	• Remove and clean the filter.
	• Suction/delivery valves are clogged or worn out.	• Contact an authorised service center.
	• Pressure regulator valve set on minimum.	• Increase the pressure by screwing in the pressure regulator knob.
	• The nozzle is worn out.	• Replace the nozzle.
Fluctuating pressure	• The nozzle is either clogged or deformed.	• Dismantle the lance and clean the nozzle with the pin provided.
	• Suction/delivery valves are clogged or worn out.	• Contact an authorised service center.
	• Worn gaskets.	• Contact an authorised service center.
	• Pressure regulator valve gaskets worn.	• Contact an authorised service center.
	• The filter is dirty.	• Remove and clean the filter.
Water leak from appliance	• Water leaking from the pump.	• Contact an authorised service center.
	• Leaks from hose reel (if present).	• Tighten the unions; if the problem persists, contact an authorised service center.
	• Leaks from pressure regulator valve.	• Contact an authorised service center.
	• Leaks from water unions.	• Tighten the unions; if the problem persists, contact an authorised service center.
Abnormal noise	• Suction/delivery valves are clogged or worn out.	• Contact an authorised service center.
	• Worn bearings.	• Contact an authorised service center.
	• The filter is dirty.	• Remove and clean the filter.
	• Extreme water temperature.	• Bring the temperature under 50°.
Water is in the oil	• O-rings (water and oil) are worn out.	• Contact an authorised service center.
The machine does not suck wax or detergent	• Tanks empty.	• Replace oil with double frequency that is suggested.
		• Fill the tanks.
	• Rod knob on high pressure position.	• Turn it to the low pressure position.
	• Disconnected pipe in detergent suction system.	• Return it into position.

WARRANTY CONDITIONS

All our machines are subjected to strict tests and are covered against manufacturing defects in accordance with applicable regulations. The warranty is effective from the date of purchase.

The following are not included in the warranty:

- Parts subject to normal wear. – Rubber parts, nozzles, hoses, filters and other consumable parts and accessories.
- Accidental damage, caused by transport, neglect or inadequate treatment, incorrect or improper use and installation failing
- The warranty shall not cover any cleaning operations to which the operative components may be subjected, such as clogged nozzles and filter blocked due to limestone.

DISPOSAL



As the owner of electrical or electronic equipment, the law (in accordance with the EU Directive 2002/96/EC of 27 January 2003 on waste from electrical and electronic equipment and the national laws of the EU Member States that have implemented this Directive) prohibits you from disposing of this product or its electrical / electronic accessories as municipal solid waste and obliges you to make use of the appropriate waste collection facilities.

The product can be disposed of by returning it to the distributor when a new product is purchased. The new product must be equivalent to that being disposed of.

Disposing of the product in the environment can cause great harm to the environment itself and human health.

The symbol in the figure indicates the urban waste containers and it is strictly prohibited to dispose of the equipment in these containers. Non-compliance with the regulations stipulated in the Directive 2002/96/EC and the decrees implemented in the various EU Member States is administratively punishable.



EC Declaration of Conformity

Document number: dh-15.4

We hereby declare that the machine described below complies with the relevant basic safety and health requirements of the EU Directives, both in its basic design and construction as well as in the version put into circulation by us. This declaration shall cease to be valid if the machine is modified without our prior approval.

Manufacturer or representative: COMBIJET MANUFACTURING LTD

Address: Agios Athanasios Industrial Estate, Gonia Falea & Koutsoventi, 4101 Limassol, Cyprus

Brand name or trade mark:

Product type: High Pressure Water Blaster

Product designation: Heavy Duty Application

The designated product(s) is (are) in conformity with the provisions of the following European Directives.

☒	2006/95/EC and amendments	Directive of the European Parliament and of the Council of 12 December 2006 on the harmonisation of the laws of Member States relating to electrical equipment designed for use within certain voltage limit.
☒	2004/108/EC and amendments	Directive of the European Parliament and of the Council of 15 September 2004 on the approximation of the laws of the Member States relating to electromagnetic compatibility
☒	2000/14/EC	Directive of the European Parliament and of the Council of 08 May 2000 on the noise emission in the environment by equipment for use outdoors
☒	2006/42/EC	Directive of the European Parliament and of the Council of 17 May 2006 on machinery

Further information regarding compliance with these Directives is given in the annex which constitutes a part of this declaration.

Last two digits of the year in which the CE marking was affixed: 18

Place and date of signatures: Limassol, ____ / ____ / ____

Serial Number: _____

Signatures:

Names: Constantinos Foutas

This declaration of conformity is issued under the sole responsibility of the manufacturer. It certifies compliance with the indicated Directives but implies no warranty of properties.

Document number: dh-15.4

The conformity of the designated product(s) with the provisions of the European Directive 2006/95/EC is given by the compliance with the following European Standard(s). If not elsewhere/otherwise indicated the edition/amendment as referenced below applies.

- ☒ EN 60335-1:2012 Household and similar electrical appliances – Safety - Deals with the safety of electrical appliances and machines for household environment and commercial purpose, their rated voltage being not more than 250 V for single-phase appliances and machines and 480 V for other appliances and machines.
- ☒ EN 60335-2-79:2009 Household and similar electrical appliances – Safety - Particular requirements for high pressure cleaners and steam cleaners

The conformity of the designated product(s) with the provisions of the European Directive 2004/108/EC is given by the compliance with the following European Standard(s). If not elsewhere/otherwise indicated the edition/amendment as referenced below applies.

- ☒ EN 1:2006+A1:2009 55014– Electromagnetic compatibility. Requirements for household appliances, electric tools and similar apparatus. Emission
- ☒ EN 2:1997+A2:2008 55014– Electromagnetic compatibility - Requirements for household appliances, electric tools and similar apparatus
- ☒ EN 61000-3-2:2006+A1:2009+ A2:2009 Electromagnetic compatibility (EMC)—Part 3-2: Limits—Limits for harmonic current emissions (equipment input current ≤ 16 A per phase)
- ☒ EN 61000-3-3:2008 Electromagnetic compatibility (EMC)—Part 3-3: Limits—Limitation of voltage changes, voltage fluctuations and flicker in public low voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subjected to conditional connection
- ☒ EN 62233:2008 Measurement methods for electromagnetic fields of household appliances and similar apparatus with regard to human exposure
- ☒ EN 61439-1:2011 Low-voltage switchgear and control gear assemblies

The conformity of the designated product(s) with the provisions of the European Directive 2000/14/EC is given by the compliance with the following European Standard(s). If not elsewhere/otherwise indicated the edition/amendment as referenced below applies.

Applied conformity evaluation method

2000/14/EC: Appendix V

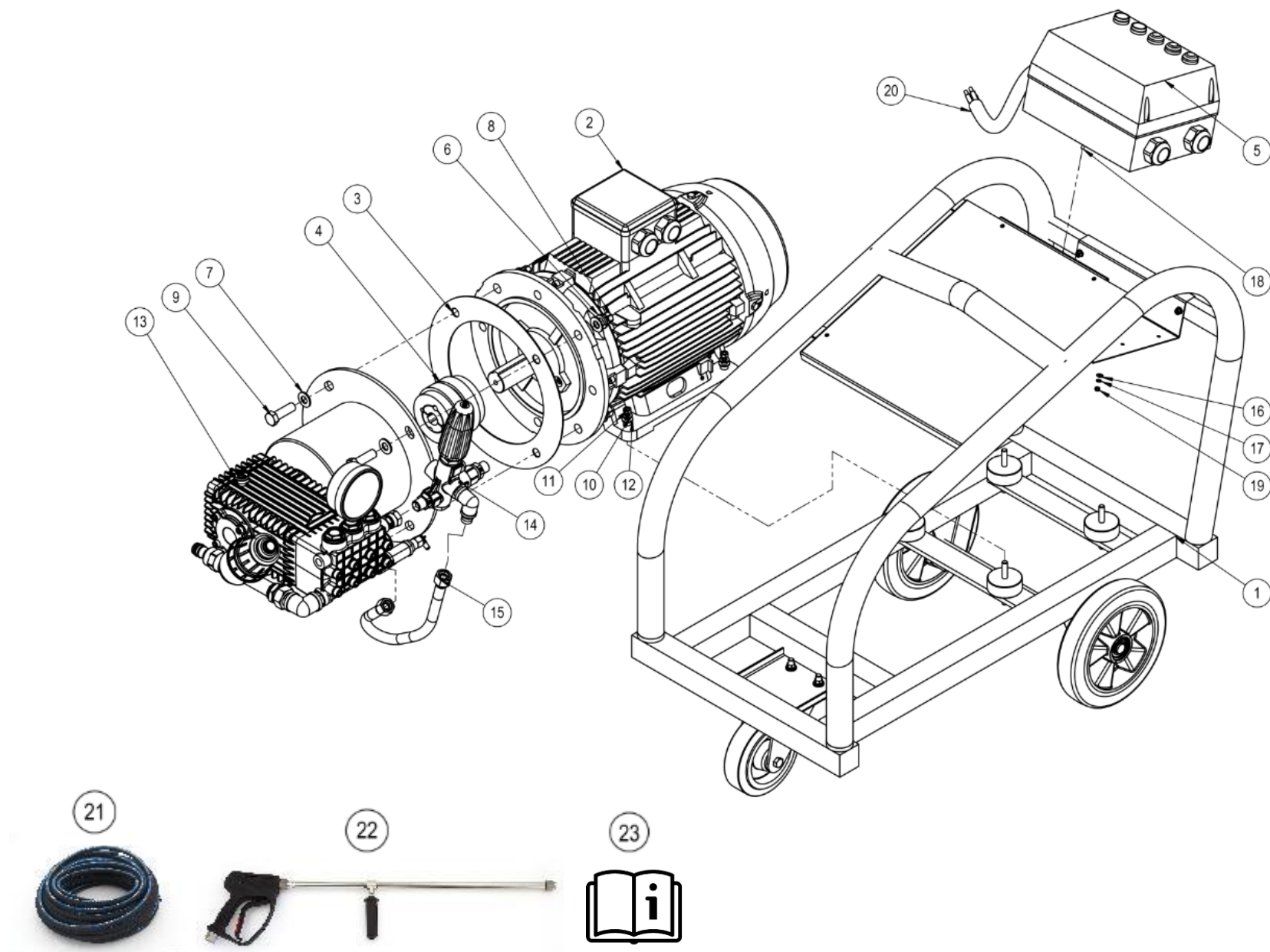
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Applied conformity evaluation method

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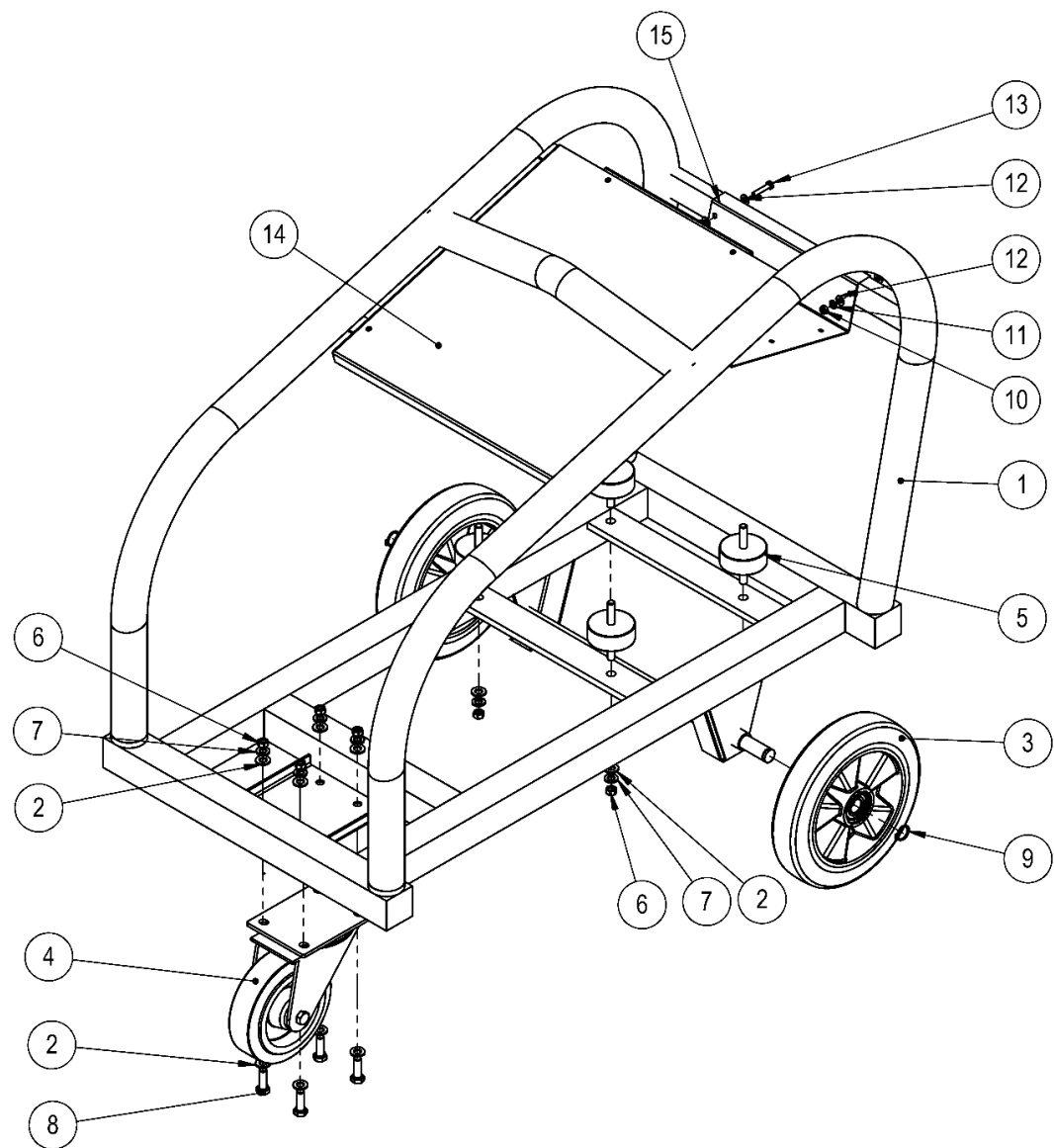
CHASSIS SPARE PARTS ASSEMBLY



CHASSIS SPARE PARTS LIST

POS.	PART NO.	DESCRIPTION	QTY.
1	CJL30949	COMPLETE STAINLESS-STEEL FRAME (REF. pg. 28)	1
2	CJL30998	15KW ELECTRIC MOTOR	1
3	CJL40176	HOUSING WASHER (OPTIONAL)	1
4	CJL40175	COMPLETE COUPLING (REF.pg.30)	1
5	CJL30932	COMPLETE ELECTRIC BOX (REF. pg.47)	1
6	CJL160SW	M16 SPLIT LOCK WASHER	4
7	CJL160FW	M16 FLAT WASHER	8
8	CJL1600N	M16 NUT	4
9	CJL1650B	BOLT M16X50	4
10	CJL100SW	M10 SPLIT LOCK WASHER	4
11	CJL1000N	M10 NUT	4
12	CJL100FW	M10 FLAT WASHER	4
13	CJL30988	HP PUMP 500BAR (REF. pg. 32,34)	1
14	CJL30377	PRESSURE REGULATING VALVE (REF. pg. 36,38)	1
15	CJL34966	RETURN HOSE ½" WITH FITTINGS	1
16	CJL500FW	M5 FLAT WASHER	4
17	CJL500SW	M5 SPLIT LOCK WASHER	2
18	CJL0525B	BOLT M5X25	2
19	CJL500SN	M5 SAFETY NUT	2
20	CJL30962	PVC CABLE 4G X 6mm ² Length: 10m	1
21	CJL30037	HOSE 500BAR 20M LENGTH WITH ½" FITTINGS	1
22	CJL30904	GUN COMPLETE WITH LANCE (REF. pg. 40,42)	1
23	CJL30901	JE60-500 INSTRUCTION MANUAL	1

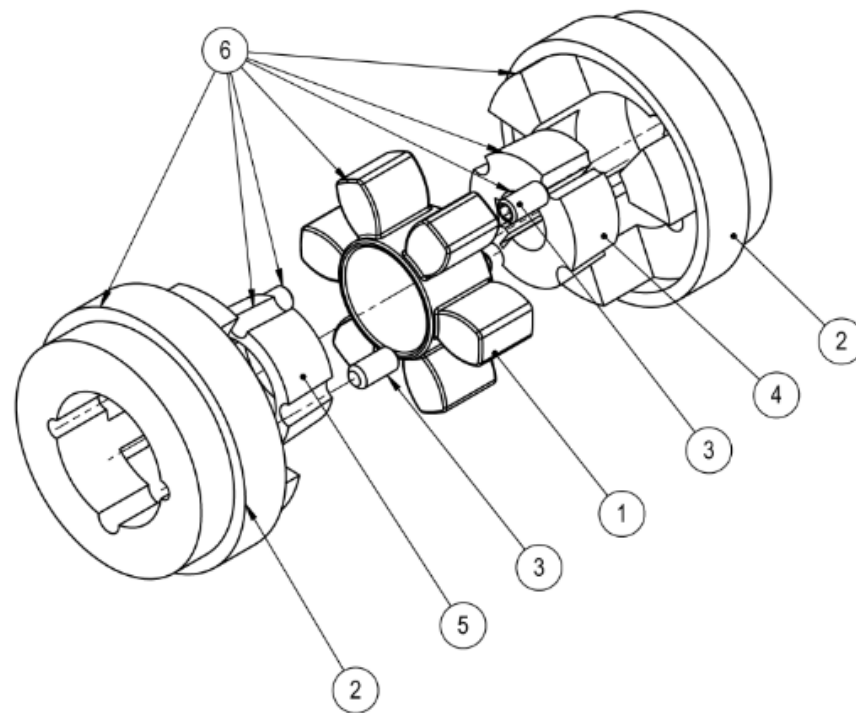
FRAME SPARE PARTS ASSEMBLY



FRAME SPARE PARTS ASSEMBLY LIST

POS.	PART NO.	DESCRIPTION	QTY.
1	CJL30949	STAINLESS STEEL FRAME	1
2	CJL100FW	M10 WASHER	12
3	CJL30973	WHEELS+BLOCK+COVER KIT X LKX'4-NPX'4	2
4	CJL30917	CASTOR WHEEL WITH BRAKE SYSTEM D:150X40	1
5	CJL30826	MOTOR RUBBER MOUNTS OD:65X25 M10X35	4
6	CJL1000N	M10 NUT	8
7	CJL100SW	M10 SPLIT LOCK WASHER	8
8	CJL01030B	M10X30	4
9	CJL0191CC	CIRCLIP 19X1	2
10	CJL6000N	M6 SAFETY NUT	2
11	CJL600SW	M6 SPLIT LOCK WASHER	2
12	CJL600FW	M6 FLAT WASHER	4
13	CJL00630B	BOLT M6X30	2
14	CJL30948	MACHINE STEEL COVER (RAL 5017)	1
15	CJL30945	STEEL BASE FOR ELECTRIC BOX (RAL5017)	1

COUPLING SPARE PARTS ASSEMBLY



COUPLING SPARE PARTS LIST

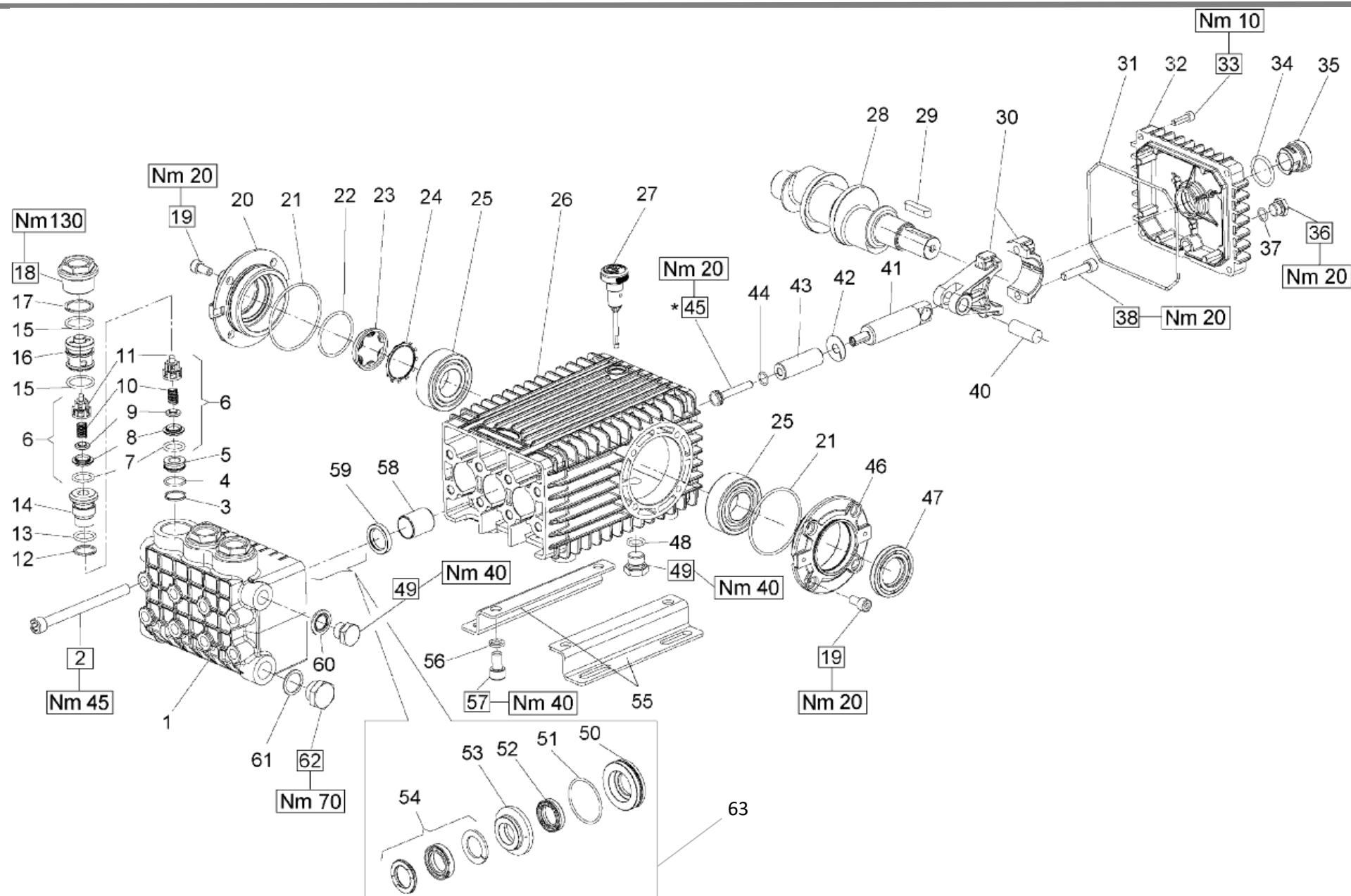
POS.	PART NO.	DESCRIPTION	QTY.
1	CJL34698	RUBBER ELEMENT	1
2	CJL34697	COUPLING HALVES	2
3	CJL40162	SCREW M10X16 ALLEN HEADLESS	4
4	CJL34699	PUMP SHAFT COLLAR ϕ 25mm	1
5	CJL34700	MOTOR SHAFT COLLAR ϕ 42mm	1
6	CJL40175	COMPLETE COUPLING	1



PUMP PARTS LIST

POS.	PART NO.	DESCRIPTION	QTY.
1	CJL800SW	M8 SPLIT LOCK WASHER	4
2	CJL800FW	M8 FLAT WASHER	4
3	CJL00830B	BOLT M8X30	4
4	CJL40162	M10X35 HEADLESS ALLEN SCREW	2
5	CJL34699	SHAFT COLLAR FOR PUMP	1
6	CJL34697	COUPLING HALVE	1
7	CJL34158	THERMOVALVE 3/8" M	1
8	CJLO12W0110	REDUCER 1/2" M TO 3/8" F	1
9	CJL30066	ADAPTOR SWIVEL 3/8" M TO 1/2" F	1
10	CJL30383	SEALING WASHER DT6	1
11	CJTE1F08F08	FEMALE TEE 1/2"	1
12	CJNP0800000	NIPPLE 1/2"	2
13	CJL30401	MALE BARBED STARTER 3/4"	1
14	CJQC12M12F	QUICK COUPLER M/F 3/4"	1
15	CJEL1F12F12	FEMALE ELBOW 3/4"	1
16	CJTE1F12000	FEMALE TEE 3_4"	1
17	CJL33950	INLET CUT OFF SENSOR _ PART 2140051	
18	CJL30304	WATER FILTER COMPLETE WITH CARTRIDGE	1
19	CJCUPBRSD25F	Express brass hose coupling female	1
20	CJCUPBRSD25M	Express brass hose coupling male	1
21	CJL40138	FILTER MESH CARTRIDGE 80micron	1
22	CJL30988	PUMP	1
23	CJL30950	HP GAUGE 1/2"	1
24	CJL40119	COUPLING HOUSING	1

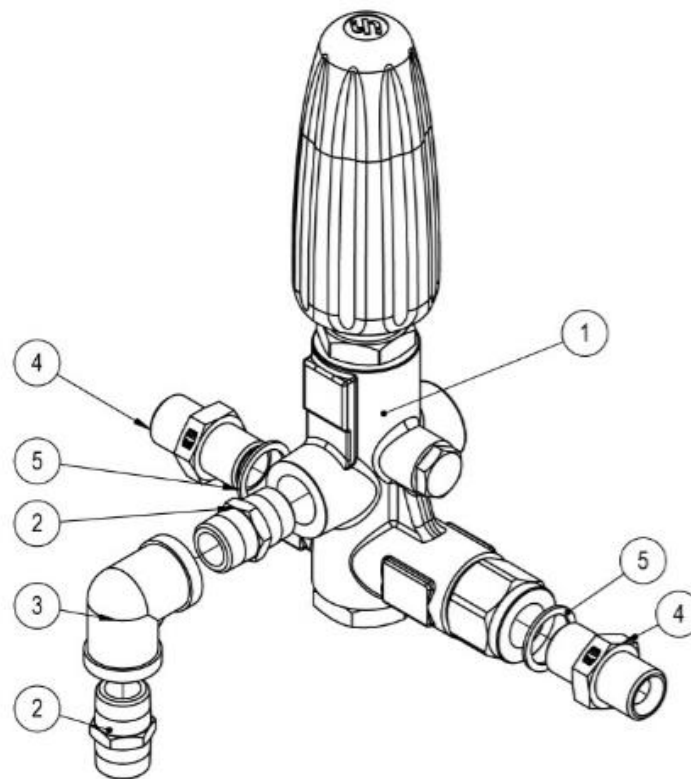
PUMP SPARE PARTS ASSEMBLY



PUMP SPARE PARTS ASSEMBLY/LIST

POS.	PART NO.	DESCRIPTION	QTY.	POS.	PART NO.	DESCRIPTION	QTY.	POS.	PART NO.	DESCRIPTION	QTY.
*	CJL30988	PUMP	1	24	CJL34491	SNAP RING	1	50	CJL34435	BACK RING	3
1	CJL34112	PUMP HEAD	1	25	CJL34390	ROLLER BEARINGS	2	51	CJL34403	O-RING Ø 34.65x1.78 (2137)	3
2	CJL34385	SCREW UNI 5931 M10x120	8	26	CJL34330	PUMP	1	52	CJL34456	RING SEAL 16x24x6.5 & 18x24x6.5	3
3	CJL34441	RINGx19.5x22x1.5	3	27	CJL34467	OIL FILLER CAP	1	53	CJL34344	MIDDLE RING	3
4	CJL34483	O-RING 18.77x1.78 Ø (2075)	3	28	CJL34305	CRANK SHAFT	1	54	CJL34313	RING SEAL Ø 16x26x9.5	3
5	CJL34322	SPACER	3	29	CJL34439	KEY	1	55	*	PUMP SUPPORTS	2
6	CJL30802	REPAIR KIT – SUCTION/DELIVERY VALVES	6	30	CJL30807	COMPLETE CONNECTING ROD	3	56	CJL34W08	Washer Ø 10.2 DIN 7980 (Optional)	4
7	CJL34347	O-RING 17.13x2.62 Ø (3068)	6	31	CJL34460	O-RING Ø 133.02x2.62 (3525)	1	57	CJL34408	Piston Bolt M10X18 UNI 5931 (Optional)	4
8	CJL34372	VALVE SEAT	6	32	CJL34387	BACK COVER	1	58	CJL34326	BUSHING (Optional)	3
9	CJL34425	BALL VALVE	6	33	CJL34333	SCREW M6x20	4	59	CJL30825	REPAIR KIT - OIL SEAL KIT	3
10	CJL34400	SPRING 9.4x14.8	6	34	CJL34307	O-RING 26.58 x 3.53	1	60	CJL34386	SEALING WASHER G3/8" Ø 15x27x2	1
11	CJL34479	VALVE GUIDE	6	35	CJL34367	OIL	1	61	CJL34370	PLUG G1/2"x10	1
12	CJL34488	RING 18.7X23X1.5	3	36	CJL34448	PLUG	1	62	CJL34346	Ø 21.5x27x1.5	1
13	CJL34392	O-RING	3	37	CJL34496	O-RING Ø 10.82x1.78	1	63	CJL30803	REPAIR KIT – COMPLETE SEAL KIT 16MM	1
14	CJL34350	BUSHING	3	38	CJL34415	SCREW M8x35	6				
15	CJL34316	O-RING	6	40	CJL34343	PIN					
16	CJL34355	PLUG	3	41	CJL34424	PISTON GUIDE	3				
17	CJL34422	RING	3	42	CJL34970	RING D.10x28x0.5	3				
18	CJL34311	VALVE CAP Normal	2	43	CJL30811	PISTON D.16x54	3				
18A	CJL34T11	VALVE CAP Threaded	1	44	CJL34380	O-RING 10.82x1.78	3				
19	CJL34310	SCREW M8x16	8	45	CJL34969	PISTON SCREW	1				
20	CJL34421	SIDE COVER	1	46	CJL34421	SIDE COVER	1				
21	CJL34445	O-RING 67.95x2.62	2	47	CJL34488	RING 18.7X23X1.5	3				
22	CJL34472	O-RING 39.34x2.62	1	48	CJL34358	O-RING Ø 13.95x2.62 (3056)	1				
23	CJL34499	OIL LEVEL	1	49	CJL34375	PLUG G3/8"x13	2				

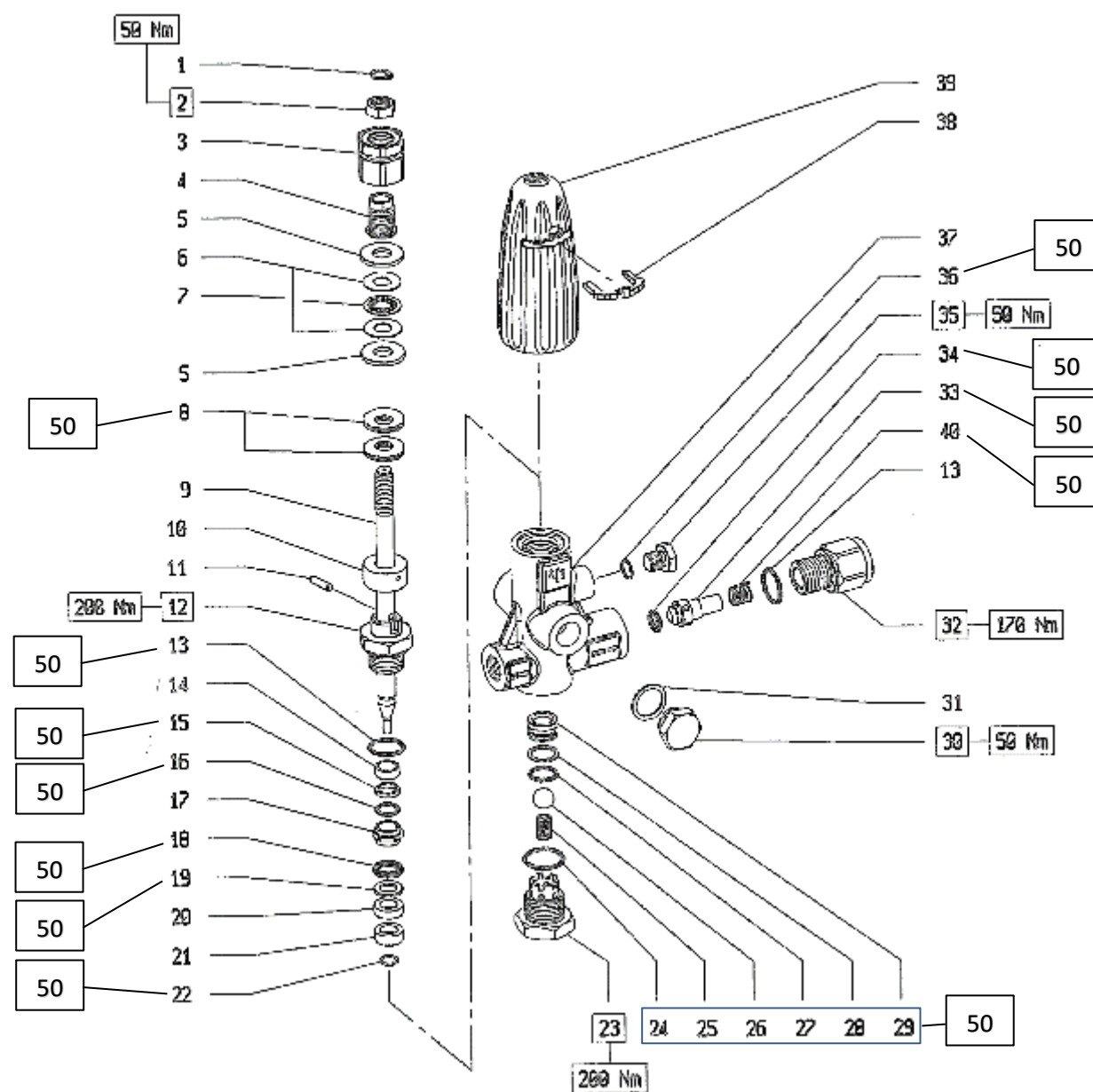
PRESSURE REGULATING VALVE FITTINGS SPARE PARTS ASSEMBLY



PRESSURE REGULATING VALVE FITTINGS SPARE PARTS ASSEMBLY LIST

POS.	PART NO.	DESCRIPTION	QTY.
1	CJL30377	PRESSURE REGULATING VALVE	1
2	CJNP0800000	NIPPLE ½"	2
3	CJEB08F08F	FEMALE ELBOW ½"	1
4	CJ20122364	S/S NIPPLE ½"	2
5	CJL30385	SEALING WASHER DT8	2

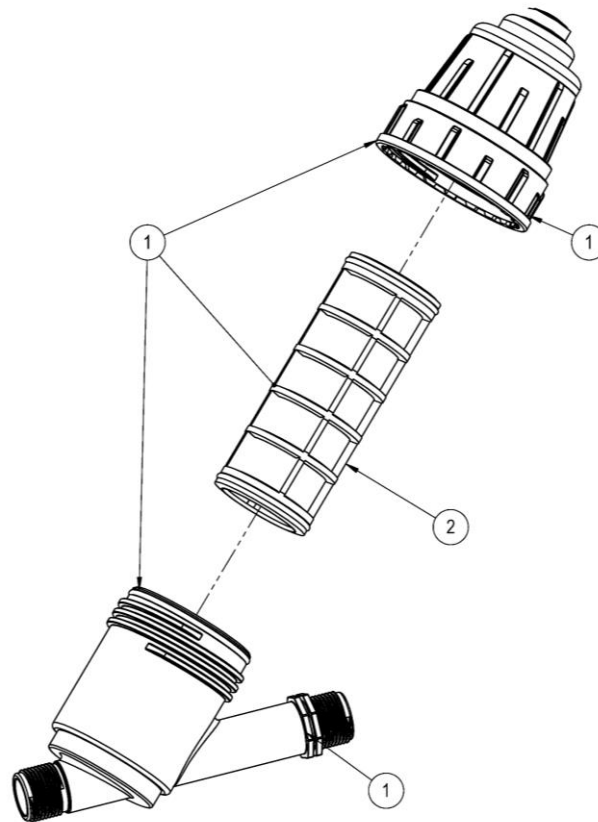
PRESSURE REGULATING VALVE SPARE PARTS ASSEMBLY



PRESSURE REGULATING VALVE SPARE PARTS LIST

POS.	PART NO.	DESCRIPTION	QTY.	POS.	PART NO.	DESCRIPTION	QTY.
-	CJL30377	PRESSURE REGULATING VALVE COMPLETE	1	22	CJL34352	RING Ø 1212	1
1	CJL34490	SNAP RING	1	23	CJL34453	CAP M 27x1.5	1
2	CJL34351	NUT M12x1.25x19	1	24	CJL34480	Ø 23.47x2.62	1
3	CJL34377	PRESSURE REGULATING RING	1	25	CJL34404	SPRING Ø 9.3x15.5 INOX	1
4	CJL34304	SCREW FOR ADJUSTING MAX. PRESSURE	1	26	CJL34447	BALL 9/16 STAINLESS	1
5	CJL34366	WASHER	2	27	CJL34334	OR Ø15.08X2.62	1
6	CJL34388	Ø 12x26x1	2	28	CJL34301	RING	1
7	CJL34416	ROLLER BEARING Ø12x26x2	1	29	CJL34428	VALVE SEAT	1
8	CJL34495	WASHER Ø 28x12.2x1.5	30	30	CJL34484	PLUG G ½ "x 10 INOX	1
9	CJL34470	VALVE HAFT	1	31	CJL34359	RING 21.5x27x1.5	1
10	CJL34378	BUSHING SUPPORT SPRINGS	1	32	CJL34105	NIPPLE G1/2" F	1
11	CJL34331	PLUG CIL. Ø 5x18	1	33	CJL34432	VALVE	1
12	CJL34363	SHAFT BUSHING GUIDE	1	34	CJL34452	OR Ø 9.12x3.53	1
13	CJL34393	OR Ø 20.63x2.62	2	35	CJL34361	PLUG G1/4"x13 INOX	1
14	CJL34468	BUSH GUIDE Ø 16.5	1	36	CJL34327	OR Ø 10.82x1.78	1
15	CJL34493	O-RING	1	37	CJL34431	BODY	1
16	CJL34366	O-RING 11.91x2.92	1	38	CJL34498	FORK	1
17	CJL34302	SPACER	1	39	CJL34323	KNOB	1
18	CJL34381	O-RING Ø 12x4.5	1	40	CJL34405	SPRING Ø 12.7x15 INOX	1
20	CJL34482	BUSH GUIDE Ø 20	1	50	CJL30822	REPAIR KIT - REGULATING VALVE	1
21	CJL34357	LOCKING BUSH	1				

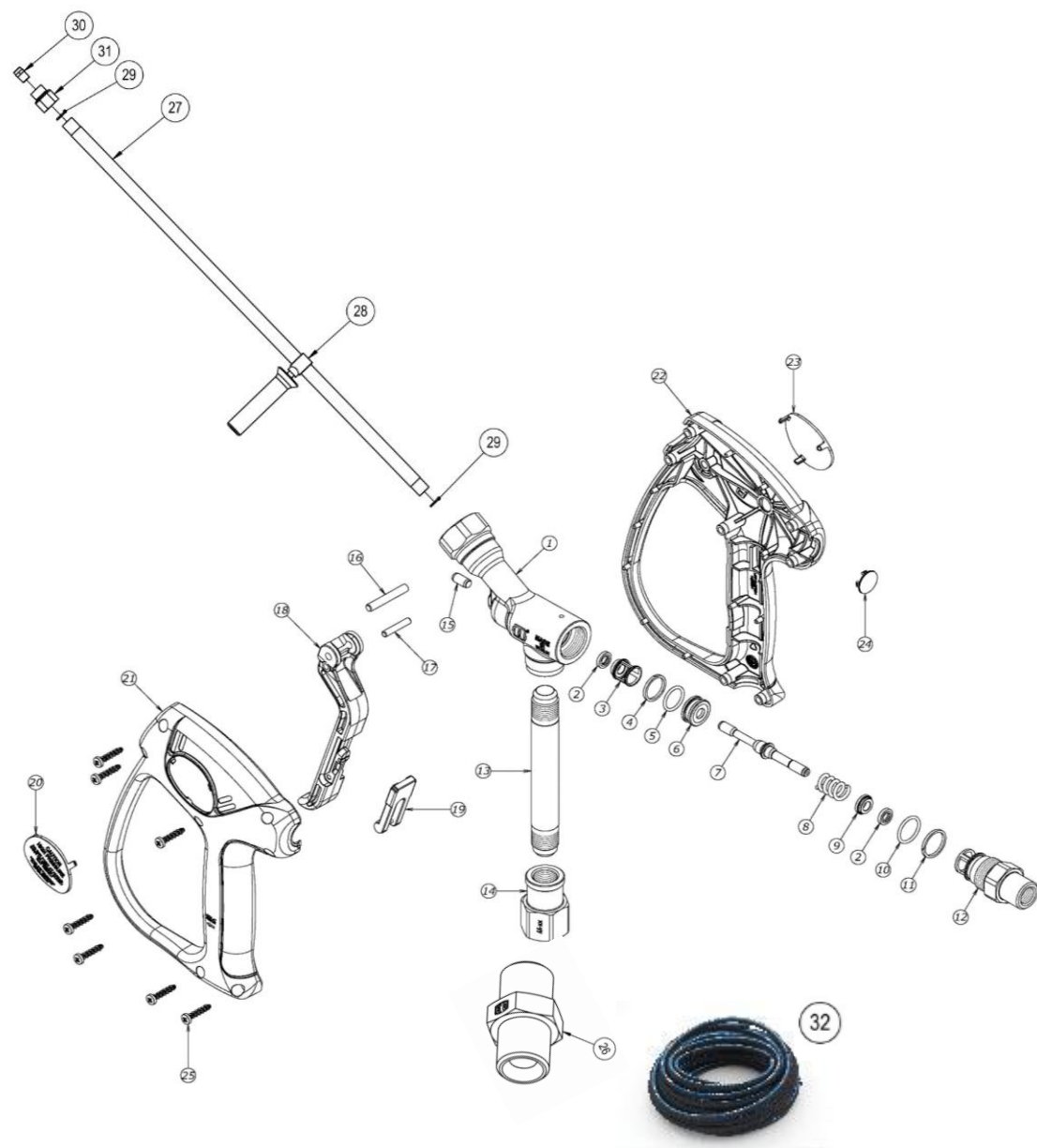
WATER FILTRATION SYSTEM



WATER FILTRATION SYSTEM

POS.	PART NO.	DESCRIPTION	QTY.
1	CJL40175	FILTER COMPLETE ¾"	1
2	CJL40138	FILTER MESH CARTRIDGE 80micron	1

DRY SHUT GUN SPARE PARTS ASSEMBLY



DRY SHUT GUN SPARE PARTS LIST

POS.	PART NO.	DESCRIPTION	QTY.
*	CJL30904	GUN COMPLETE (POS.1-27)	1
1	CJL34825	NICKEL BODY G1/2" F	1
2	CJL34826	HD SLIPPER NCR Ø61N + N-103+OR	2
3	CJL34827	CONVEYING SPACER	1
4	CJL34828	A.A 15,20X18,00X1,70 TBT	1
5	CJL34829	GUOR 1.78X14 2056 BLACK	1
6	CJL34830	SEAT	1
7	CJL34831	STEM	1
8	CJL34832	SPRING 2.2X9.8X27.0	1
9	CJL34833	SPACER	1
10	CJL34834	GUOR 1.78X15.6 2062 BLACK	1
11	CJL34835	A.A 16.70X19.5X1.70 TBT	1
12	CJL34836	FITTING	1
13	CJL34837	INOX TUBE Ø17X3 L126 G3/8"	1
14	CJL34838	INLET FITTING G1/2" F MARC	1
15	CJL34839	BRASS PIN Ø6X12.00	1
16	CJL34840	LEVER PIN D.5X33 IX	1
17	CJL34841	LEVER PIN D.4X22 INOX	1
18	CJL34842	TRIGGER BLACK	1
19	CJL34843	SAFETY - RED	1
20	CJL34844	LEFT PLATE	1
21	CJL34845	LEFT SHELL BLACK	1
22	CJL34846	RIGHT SHELL BLACK	1
23	CJL34847	SAFETY - RIGHT	1
24	CJL34848	PLASTIC PLUG	1
25	CJL34849	SCREW M4X19 UNI9707	7
26	CJ20122364	S/S NIPPLE M½"	1

POS.	PART NO.	DESCRIPTION	QTY.
27	CJL34153	LANCE 800mm	1
27	CJL90G0003	LANCE 400mm (Optional)	1
27	CJL90G0125	LANCE 1200mm (Optional)	1
28	CJL30487	SIDE HANDLE	1
29	CJL30501	COPPER SEALING WASHER	2
30	CJL32100	STRAIGHT JET 0° ¼" (OPTIONAL)	1
30	CJ20141010	FAN JET NOZZLE SIZE 030 (50Hz)	1
30	CJ20122072	FAN JET NOZZLE SIZE 035 (60Hz)	1
31	CJL30528	NOZZLE HOLDER	1
32	CJL30037	HOSE 500BAR 20M LENGTH WITH ½" FITTINGS	1
KIT 1	CJL34822	REP KIT 1- SEALS 500 BAR (AT POS. 2,4,5,6,7,8,10,11)	1
KIT 2	CJL34922	REP KIT 2- TRIGGER GUN 500 BAR (AT POS. 15,16,17,18,19)	1

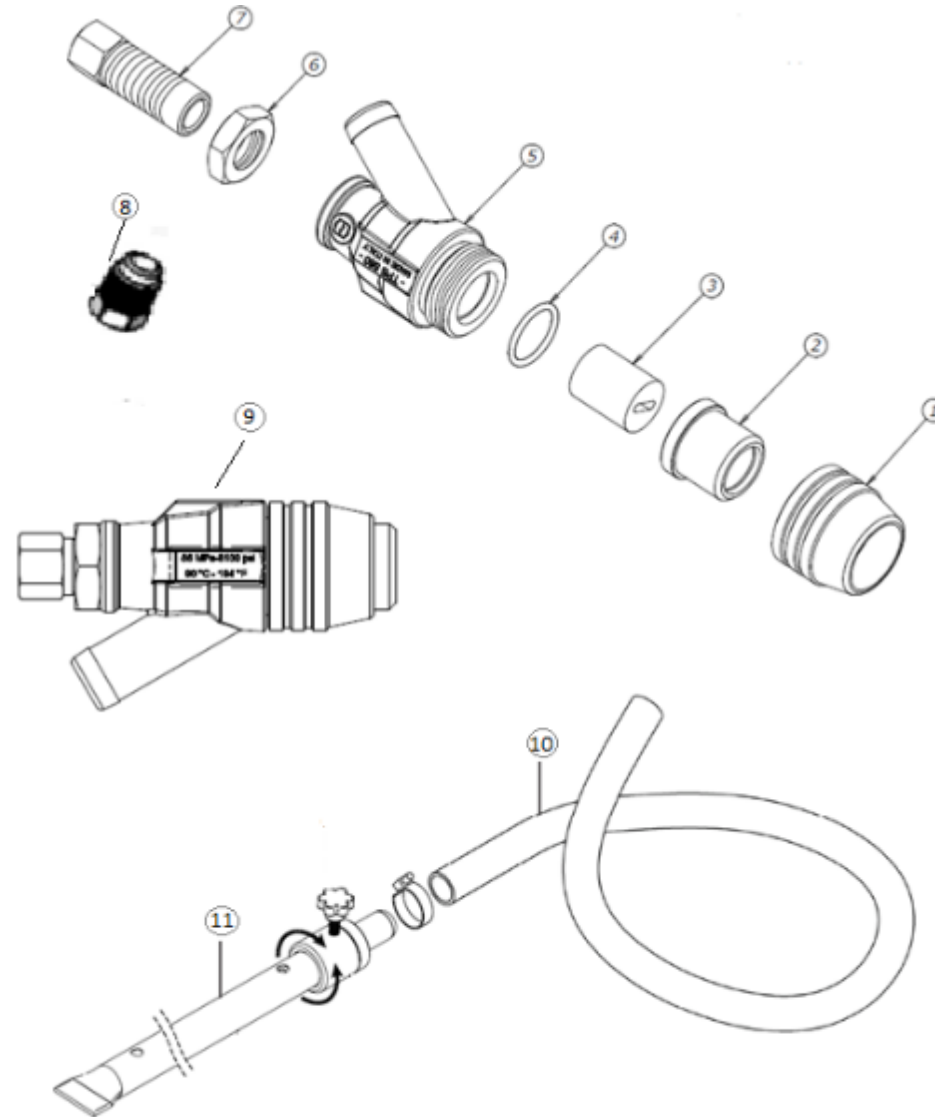
ROTATING NOZZLE



ROTATING NOZZLE SPARE PARTS LIST

POS.	PART NO.	DESCRIPTION	QTY.
1	CJL33503	ROTATING NOZZLE 0.030"	1
	CJL31222	ROTATING NOZZLE 0.035"	1
	CJL31304	ROTATING NOZZLE 0.040"	1
	CJL35121	ROTATING NOZZLE 0.045"	1
	CJL31305	ROTATING NOZZLE 0.050"	1
	CJL30355	ROTATING NOZZLE 0.055"	1
2	CJL33513	REPAIR KIT FOR ROTATING NOZZLE 0.030"	1
	CJL35222	REPAIR KIT FOR ROTATING NOZZLE 0.035"	1
	CJL31340	REPAIR KIT FOR ROTATING NOZZLE 0.040"	1
	CJL35152	REPAIR KIT FOR ROTATING NOZZLE 0.045"	1
	CJL31355	REPAIR KIT FOR ROTATING NOZZLE 0.050"	1
	CJL36355	REPAIR KIT FOR ROTATING NOZZLE 0.055"	1
-	CJ20122301	NIPPLE HP SS ¼" FOR ROTATING NOZZLE	1

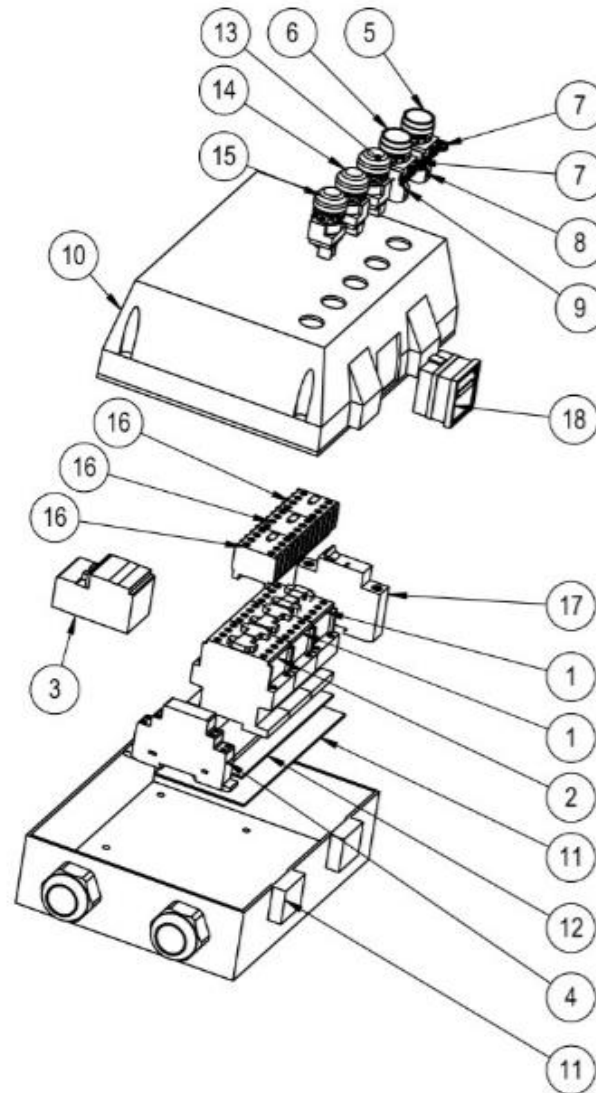
WET SAND-BLASTING ACCESSORY ASSEMBLY



WET SAND-BLASTING ACCESSORY LIST

POS.	PART NO.	DESCRIPTION	QTY.
*	CJL30320	WET SAND-BLASTING ACCESSORY- COMPLETE 50HZ (POS.8-11)	
*	CJL30360	WET SAND-BLASTING ACCESSORY- COMPLETE 60HZ (POS.8-11)	
1	CJL30312	RING	1
2	CJL30315	NOZZLE HOLDER	1
3	CJL30317	NOZZLE HEAD	1
4	CJL30331	O-RING 13223.81X2,62	1
5	CJL30338	BODY	1
6	CJL30344	NUT SS M20x1,5	1
7	CJL30348	CONNECTOR	1
8	CJ20141010	WATER NOZZLE - FANJET SIZE 0.30	1
8	CJ20122072	WATER NOZZLE - FANJET SIZE 0.35	1
9	CJL30334	WET SAND-BLASTING HEAD COMPLETE (POS1-8)	1
10	CJL30328	HOSE – 10m	1
11	CJL30335	ADJUSTABLE SUCTION PROBE	1

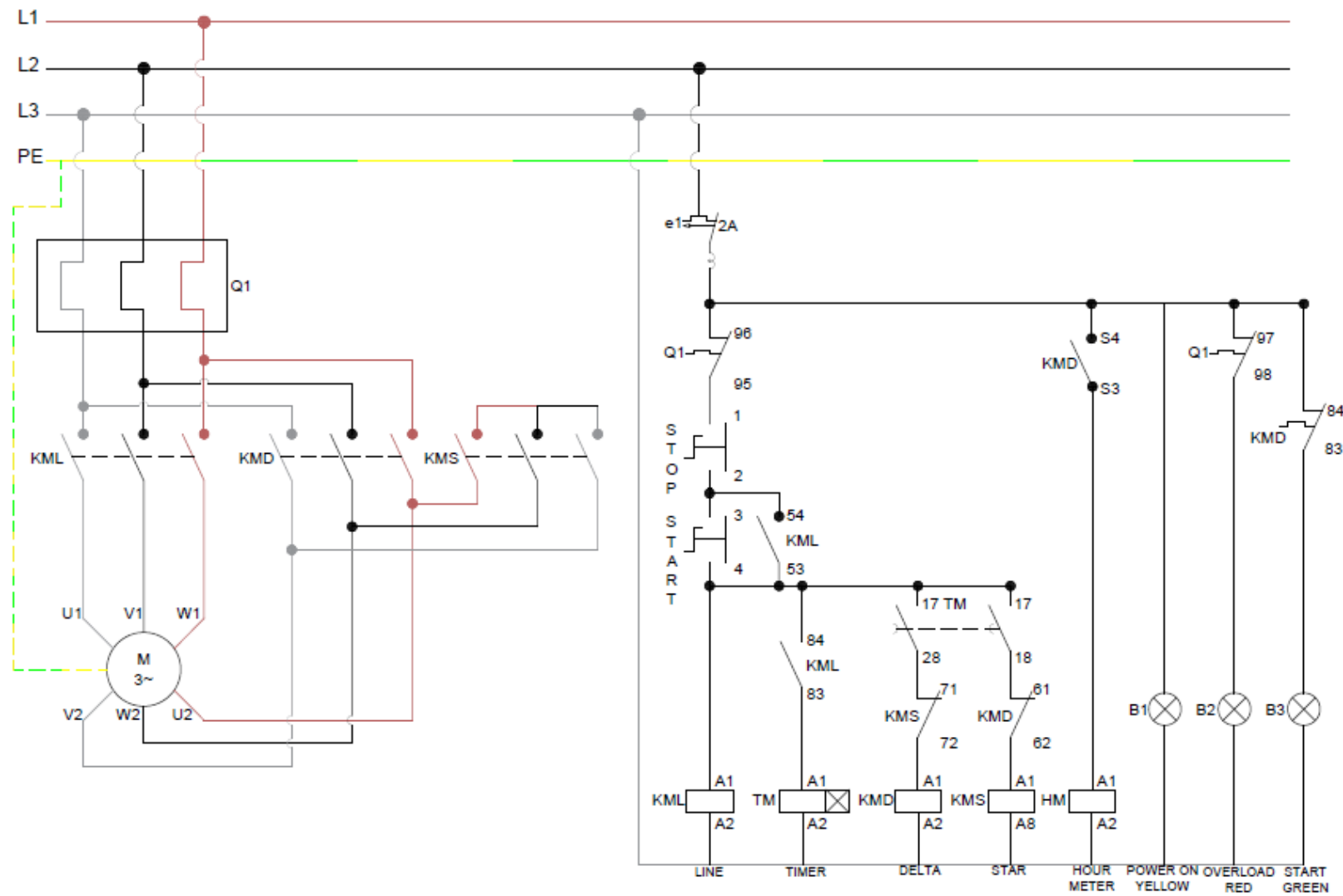
ELECTRICAL PANEL SPARE PARTS ASSEMBLY



ELECTRICAL PANEL SPARE PARTS LIST

POS.	PART NO.	DESCRIPTION	QTY.
1	CJL30861	3P CONTACTOR 1NO 18A AC3 400/50-60	2
2	CJL30869	3P CONTACTOR 1NO 12A AC3 400/50-60	1
3	CJL90E096	THERMAL OVERLOAD RELAY 13-18A	1
4	CJL90E0144	TIME RELAY S/D 380/440V 50/60	1
5	CJL10E21	PUSH BUTTON-GREEN	1
6	CJL10E22	PUSH BUTTON-RED	1
7	CJL10E17	MOUNTING ADAPTER	2
8	CJL10E06	AUXILIARY CONTACT-NO	1
9	CJL10E08	AUXILIARY CONTACT-NC	1
10	CJL30311	ENCLOSURE WITHOUT BUTTONS	1
11	CJL30322	REAR METAL PLATE FOR M3	1
12	CJL30804	LED INTEGRATED MONOBLOCK PILOT LIGHT GREEN	1
13	CJL30818	LED INTEGRATED MONOBLOCK PILOT LIGHT RED	1
14	CJL30814	LED INTEGRATED MONOBLOCK PILOT LIGHT YELLOW	1
15	CJL10E05	AUX.CONTACT 2NO+2NC CENTRE MOUNTING	3
16	CJL10E18	MCB	1
17	CJL90E0091	HOUR METER 50HZ 440VAC	1

ELECTRICAL SCHEMATIC



OPTIONAL PROTECTIVE EQUIPMENT



A/A	PART NO.	DESCRIPTION	QTY.
1	CJL30393	HELMET COMPLETE - WITH VISOR, EARMUFFS AND ADJUSTMENT	OPTIONAL
2	CJL34180	PROTECTIVE APRON	OPTIONAL
3	CJL30301	PROTECTIVE OVERALL	OPTIONAL
4	CJL30323	PROTECTIVE BOOTS	OPTIONAL
5	CJL30378	PROTECTIVE GLOVES	OPTIONAL



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